

Comparing Public–Private Partnerships and Traditional Public Procurement: Efficiency vs. Flexibility

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ABSTRACT *Public–private partnerships (PPPs) have become an increasingly popular way for governments to procure for their citizens certain public services. Supporters argue that the private sector can provide services more efficiently while critics complain that the long-term contracts involved reduce governments’ ability to adapt to changing needs. This paper shows that the optimal choice between a PPP and traditional public procurement depends on a number of factors, including the likelihood that changes will be necessary, the productivity of non-contractible effort exerted by private sector partners, and the bargaining power of government vis-à-vis private parties. It also shows that this choice may depend on whether the government’s objective is to maximize “value for money” or to maximize total social surplus.*

Keywords: public–private partnerships; efficiency vs. flexibility; contracting out

Introduction

Continuing a movement that had earlier involved the privatization of many state-owned firms, the deregulation of major industries and the more widespread use of contracting out, in the early 1990s a number of governments began to experiment with another innovation involving expanded use of the private sector in the delivery of public services: public–private partnerships (PPPs).¹ A large and diverse set of definitions of PPPs exist today, but one that conveys the main idea simply defines a PPP (optimistically) as an agreement between the government and one or more private partners (which may include the operators and the financiers) according to which the private partners deliver the service

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in such a manner that the service delivery objectives of the government are aligned with the profit objectives of the private partners and where the effectiveness of the alignment depends on a sufficient transfer of risk to the private partners.² For infrastructure projects (our focus here) this involves allocating the various tasks associated with the provision of the public services – for example, financing, design, construction, operation, maintenance – between public and private sector partners in ways that will provide the desired services at lower cost.³ Very long-term, and frequently complex, agreements typically govern the relationship between the private and public partners in infrastructure PPPs.

The increased private involvement in the provision of public sector infrastructure and related services has become a global phenomenon. Data from the *Infrastructure Journal* suggests that major PPP projects with a combined value of US\$394 billion have been undertaken worldwide since 2005.⁴ In both higher and lower income countries, common industries for this kind of private participation included those related to power, transportation, water and telecommunications. Various kinds of social infrastructure such as hospitals, prisons and schools are common as well.

Advocates of the PPP model argue that such arrangements, by giving a larger set of tasks to the private sector under competitive bidding, will unleash the superior power of competitive, for-profit enterprises to minimize costs and to find innovative approaches to addressing social needs. Critics of the PPP model have pointed to a number of potential problems with this model of public sector procurement. One of these – our focus here – is the loss of flexibility that comes with the long-lived contractual obligations governments must respect when changing circumstances may require significant changes in the way the public service is provided.

This suggests a trade-off faced by governments choosing whether to proceed with a project as a PPP or by traditional public procurement: while PPPs might have the potential to generate substantial productive efficiencies, they may limit a government's ability to react to changing demands for the public service. This trade-off is the subject of this paper. Our model allows us to illustrate the trade-off very simply and to explore various factors which will influence that trade-off. We can show that the optimal choice of procurement mode will depend on the exact nature of the government's objective function. In this regard, we study two commonly proffered objectives: (i) minimizing the cost to taxpayers (or users) of the service and (ii) maximizing total social surplus. We also demonstrate that the relative advantages of the PPP mode will depend on the nature of the PPP contract considered – that is, whether the private partner is compensated based on simply having completed the project or based on the actual usage of the facilities.

Some Background on PPPs

Essentially, a PPP involves contracting out at a scale and complexity well beyond what is normal for governments.⁵ As suggested, the alleged benefits of PPPs derive from their potential to apply the forces of competition and higher-powered incentives to generate higher levels of efficiency and innovation. While the experiences of governments with PPP are varied, there is evidence to suggest that the PPP model can be successful in the right circumstances.⁶

To date, a great deal of the work by economists on PPPs has been concerned with the optimality of “bundling” of the various tasks so that one private partner tackles multiple tasks (e.g. building and operating) rather than having each task performed by

a single independent private partner.⁷ While, to some researchers, it may be the bundling of tasks that defines a PPP, our focus is on the “privatization” aspect of PPPs – that they involve the outsourcing of tasks that might otherwise have been performed by public sector agencies and their employees.⁸

Critics of PPPs point out that any efficiencies coming from the increased role of the private sector come at some cost. For example, many have argued that upon entering into long-term partnership arrangements governed necessarily by complicated, detailed and still incomplete contracts, governments lose control of key aspects of service delivery that they may wish to adjust in the future. To make changes requires negotiation, and not only will the renegotiation process involve a new set of transaction/bargaining costs, it will also involve bargaining without competition on the selling side. In this sense, we say PPP agreements lack flexibility. The flexibility question strikes us as particularly important in dynamic industries such as healthcare, where the kind of hospital services appropriate today may not closely resemble the services we will be demanding in 20 or 50 years, given advances in medical knowledge and technologies.⁹ On the other hand, flexibility may be less important for other kinds of projects, such as roads and bridges.¹⁰ This paper is about how the government might trade off flexibility for efficiency – a question we believe has not been formally analyzed to this point.

A great deal of experience suggests that flexibility challenges are very real in PPP delivery. Specifically, it would appear that renegotiation is frequent and often costly. A review by Guasch (2004), for the World Bank Institute, of more than 1,000 PPP concessions granted between 1985 and 2000 in Latin America and the Caribbean reported that renegotiation “was extremely common among the concessions in the sample, occurring in 30 percent of them”.¹¹ Another review of changes made in PPP-type projects in the United Kingdom by the National Audit Office (UK) (2008, p. 4) concluded that changes undertaken in UK projects in 2006 involved extra payments to contractors of approximately £180 million. While much of this spending provided additional value to taxpayers and users, the National Audit Office (UK) (2008, p. 5) noted that “several components of the cost of changes were problematic”.¹²

While many writers have mentioned the potential costs to the public sector of lost flexibility in lengthy PPP contracts, there has been very little formal modeling to our knowledge.¹³ The paper most closely related to the questions addressed here is, in fact, not about public sector procurement. Bajari and Tadelis (2001), hereafter BT, study private sector procurement, emphasizing an example from the construction industry. As here, they emphasize the trade-off between contracts that provide maximum flexibility to incorporate changes and those that provide the strongest incentives for cost minimization. They find advantages to fixed-price contracts in terms of efficiency, but also recognize the greater flexibility in contracts of the cost-plus variety. While there are many parallels between the analysis of BT and that provided here, there are several important points of differentiation. First, our focus on the special problems associated with public sector procurement creates differences. For example, we consider different possible governmental objective functions and how they might lead to different choices of procurement mode. Second, our modeling approach is different in a number of ways. For example, we use a less “reduced form” model that, while less general in some respects, provides for a larger set of specific and intuitive comparative statics results on the importance of key parameters. Third, the source of the inefficiencies that can arise in PPP or fixed-price contracts differs in our two models. Fourth, we model the renegotiation process differently.¹⁴

Finally, we note that there is clearly a relationship between our model of costly renegotiation and models of transaction costs. Indeed, the costs of renegotiation can be viewed as an important element of the transaction costs associated with using the PPP model, and high levels of transaction costs have previously been flagged as a problem associated with the PPP model by, for example, Vining and Boardman (2008).¹⁵ As flexibility issues can, in principle, be dealt with by developing more costly (because of transaction costs) contracts, this paper can be thought of as a special case of the choice between low production costs (from private producers) and low transaction costs (by using the public sector and avoiding costly contracts).

In what follows, the next section lays out the general model, while the following section studies the two special cases, PPP and public provision, that form the core of our analysis. The article then extends the basic model in two directions. The last section concludes.

General Model

We take a fairly general approach to the problem of a government, “G” (as principal), procuring certain public services (e.g. an infrastructure “project”) from a firm, “F” (the agent). Special cases of this model will correspond to provision via public–private partnerships (denoted PPPs) and more traditionally by public agencies (PUB).¹⁶ To be clear, our model of traditional public procurement (PUB) is essentially one in which the provision of services is kept in-house by the public sector – that is, we are focusing on the degree to which the PPP model “privatizes” aspects of service delivery rather than on bundling or other potential differences between PPPs and other procurement models.

Basic Setup

The (net present value of the) gross social benefit of the project, which is neither verifiable nor contractible, is given by b_i with $b_i \in \{b_L, b_H\}$ and $b_H > b_L > 0$. The government can contract with F to deliver these services. There are in general many identical firms willing to compete to provide these services.

The firm’s cost to deliver the project depends, in part, on the non-contractible level of effort or innovation, e , chosen by the firm, and the productivity of that effort.¹⁷

$$C = K - \delta e$$

where K is the observable innate cost of the project (including cost of materials, labor, etc.) and $\delta > 0$ captures the marginal productivity of the firm’s effort. We assume that δ is the firm’s private information, and hence that it is not known by the government, *ex ante*. The monetary value of the disutility costs of this effort for the firm is $\emptyset(e)$. For simplicity, we assume that these costs can be represented by a quadratic disutility function:

$$\emptyset(e) = \frac{e^2}{2}$$

The non-verifiability of effort will generate familiar moral hazard problems.

Firms wishing to secure the contract to deliver this project will bid competitively. The firm winning the project will be paid a lump sum α by the government and then be

responsible for project costs. The firm's profits are taxed by the government at the rate $t \in [0, 1]$. We will explore the consequences of different rates of taxation but assume the level is set by forces outside the model. A firm winning the contract to deliver the project will have after-tax profits given by:

$$\pi = (1 - t)(\alpha - K + \delta e) - \mathcal{O}(e)$$

The original specification of the project comes from government and the government can be right or wrong about this specification. If the government specifies the project correctly, the social benefits will be b_H . However, there is a chance – with probability μ – that the government will have incorrectly specified the project, or that circumstances will change such that a different design is appropriate. We will refer to this as an unexpected change in the nature or level of demand for the services.¹⁸

While the change in demand is public information, we assume that because demand could have changed in so many different ways, contracts cannot practically be made contingent on changes in demand. Adapting to the new design will restore the b_H benefits. However, if the project is not changed, the lower social benefit b_L will be realized.

There are many reasons why projects may need to be redesigned or adjusted. In some cases, demand may have been badly estimated or technological changes may alter the way services should be delivered (common in healthcare, for example). While we will model this adjustment, if it is needed, as occurring before any services have been delivered, this is a modeling device meant only to capture the need for change sometime during the long life of the project.

If demand changes, efficiency dictates that the project adapt. Since this change represents a change in the contract between G and F, it must be negotiated. We model the negotiations using the familiar Nash bargaining model: the parties will bargain over the surplus created by efficiently adapting to the demand changes.¹⁹ We assume that there are real economic costs associated with switching to a new project design (this could include the renegotiation costs) and each of the two parties will bear this cost by paying an amount of $s \geq 0$.²⁰ After the contracts are settled and, if necessary, any design changes negotiated, the firm picks its level of effort, costs are incurred, contracts honored and payoffs received.

Timing

Reviewing the timing of the game, then, we have:

1. Government announces it has a project for which it wishes to receive bids.
2. Firms bid for the project, and it is awarded to the firm offering to provide it at the lowest fixed fee, α_0 .
3. Nature may move to change demand – if no change, proceed to 5.
4. If demand changes, α_0 is renegotiated (to α_1) via Nash bargaining and the design is changed, both parties incur switching costs, s .
5. F chooses level of effort, e .²¹
6. Benefits are realized and the government honors its contract.

The objective of the firm in this model is straightforward. It will make decisions to maximize its profits subject to honoring its contracts. The government's objectives are more

interesting. In much of the literature on public–private partnerships, it is argued – often just assumed – that a PPP will dominate the public procurement method in a particular case if the PPP form provides “value for money” (VFM) relative to traditional methods of procurement. Value for money in this context is typically taken to mean that services of the quality desired are provided at a lower cost to the ultimate payers – taxpayers or (in the case of tolled services) end users. This will be our starting point here: we will look for conditions under which a PPP can provide services at a lower cost to the government.

Value for money as an objective can be contrasted with the objective of maximizing total social surplus (TSS) that is often applied in cost–benefit analysis. The key differences arise due to transfers that will not matter under a TSS standard but will affect VFM. For example, firm profits will hurt VFM by pushing up the price the government pays for the project, but as simple transfers from taxpayers to firms will not affect TSS. A later section of this paper will consider how the choice between PPP and traditional methods (PUB) can depend on the government’s objective function.

Solving the Model

After solving this model for the more general case, we will focus on two specific special cases that will allow us to highlight the differences between public–private partnership and traditional public procurement. Traditional public procurement, PUB, will correspond to the case in which $t = 1$ (so that all profits revert to the government) and all bargaining power remains with the government (as it can order its employees to take actions without the need for renegotiation required with an outside contractor). We will contrast this with something more “private”, here by considering as our PPP example the case in which $t = 0$ and the bargaining weights are more equal.

We solve the model backwards, beginning with the firm’s choice of effort.

Firm’s optimal effort (e^*): Given α , F chooses a level of e to maximize its profit. This gives us the incentive compatibility constraint:

$$\max_e \pi = \max_e (1 - t)(\alpha - K + \delta e) - \emptyset(e)$$

subject to $e \geq 0$

The optimal level of effort is then given by:

$$(1 - t)\delta = \emptyset'(e^*) = e^* \quad (1)$$

Clearly, higher rates of taxation discourage effort. At the extreme, when $t = 1$, $e^* = 0$. To be clear, under our assumptions $e^* = (1 - t)\delta$ is the profit-maximizing choice of effort for either the original or a revised design, as the costs of delivering to either design are described by the same cost function and the effort decision is made after any required renegotiations.

Renegotiation (if needed): As indicated, with probability μ there is a shift in demand which requires that changes be made to project design to preserve the maximum social benefits (b_H). In such a case, if switching costs are not too large – which is assumed²² – G and F will renegotiate and sign a new contract with payment α_1 going from G to F. Under Nash bargaining, G’s threat point is: $b_L + t(\alpha_0 - K + \delta e^*) - \alpha_0$. If renegotiation fails, G gets only b_L of social benefit from the project but still pays α_0 to F as provided in the initial

contract. It also collects taxes on F's profits to reduce its net cost of the project. Correspondingly, F still receives α_0 and its optimal cost-reducing effort remains the same. Thus F's threat point will be given by:

$$(1 - t)(\alpha_0 - K + \delta e^*) - \frac{(e^*)^2}{2} = 0$$

We allow for the possibility that the government and the firm have unequal bargaining power, represented by differing bargaining weights in the Nash Product (NP). It may be, for example, that governments will have larger weights given their powers to enact and revise laws and regulations and even to rewrite contracts while shielding themselves from actions for damages. In other cases, however, sophisticated and highly motivated private partners may be able to retain higher quality advisory services to assist in their renegotiations, in the process capturing a larger share of the surplus generated.²³ To explore the implications of unequal bargaining power, suppose G has a bargaining weight λ , and F has the weight $1 - \lambda$, where $0 \leq \lambda \leq 1$.

The Nash Product will then be:

$$\begin{aligned} \text{NP} = & [(b_H + t(\alpha_1 - K + \delta e^* - s) - \alpha_1 - s) - (b_L + t(\alpha_0 - K + \delta e^*) - \alpha_0)]^\lambda \\ & * \left[\left((1 - t)(\alpha_1 - K + \delta e^* - s) - \frac{(e^*)^2}{2} \right) - \left((1 - t)(\alpha_0 - K + \delta e^*) - \frac{(e^*)^2}{2} \right) \right]^{1-\lambda} \end{aligned}$$

Maximizing this with respect to α_1 yields:

$$\alpha_1 = \frac{1 - \lambda}{1 - t} (b_H - b_L) + \frac{2\lambda - 1 - t}{1 - t} s + \alpha_0 \quad (2)$$

Initial payment (α_0): Assuming perfectly competitive bidding and no uncertainty about which design is optimal, we would expect potential private partners to bid down to the level of costs (which would be perfectly known to the firm and revealed through bidding), leaving the firm with zero profits. When we introduce uncertainty in the efficient design and the opportunity for renegotiation in the absence of competition, however, the picture is more complicated. Renegotiation under Nash bargaining, if needed, will necessarily involve moving to the new, superior design and to a readjustment upward of the fixed fee (to α_1) to share the gains of the renegotiation with the private partner. If the original contract was going to provide zero profits to the private partner (because it had been competitively bid), the renegotiated contract will necessarily generate positive profits.

Clearly, the potential for this renegotiation – and the new profits it will provide the private partners – could affect the original bidding. In a world in which private, risk-neutral, bidders could commit to honoring their promises to complete projects regardless of the subsequent profitability of those projects – perhaps by posting bonds – these prospective profits in renegotiation would translate into lower prices being bid in the initial round. The result would be that *ex ante* expected profits of bidders would still be zero – and if the design did not need to be changed, the private partner would in fact suffer losses *ex post*. However, it is

not clear that such commitment – particularly on large projects – is possible. Private partners will often be able to walk away from projects once the anticipated profits turn negative, or to at least make credible threats to walk away if subsidies are not forthcoming. This behavior is facilitated by the “special purpose vehicle” (SPV) structure that is adopted for many large PPP projects in which the SPV can simply declare bankruptcy and close down without there being any remaining claims on the SPV’s joint-venture parents.

Therefore, we invoke a limited liability constraint, familiar in the contracting literature, and assume here that the government will not accept as a winning bid any offer that would not allow the private partner to at least break even without any renegotiations.²⁴ Governments, in their scrutiny of bids, will need to be persuaded that the contract can be honored without the necessity for any renegotiations. Therefore, if we assume that a private firm has limited liability and no wealth it can commit – that is, it cannot be pushed to negative profits (or it would exit) – F will again bid up to the point where it will break even without renegotiation. The limited liability/wealth approach to the agency problem has been applied by many others as an alternative to introducing risk aversion on the part of one or both players.²⁵

We can then write the limited liability constraint as:²⁶

$$(1 - t)(\alpha_0 - K + \delta e^*) - \frac{(e^*)^2}{2} = 0$$

And the winning bid is given by:²⁷ $\alpha_0 = K - \delta e^* + \frac{(e^*)^2}{2(1-t)}$

Given that $e^* = (1 - t)\delta$, we can see that:

$$\alpha_0^* = K - \frac{\delta^2}{2}(1 - t) \quad (3)$$

This result suggests that the winning bid will be lower when tax rates are lower, because of the higher levels of effort that follow lower taxes.

Value for money (VFM): Given the costs of the contract to the government with and without renegotiation, we can easily determine the expected value for money. If there were no change in demand, the value for money would be simply given by the difference between the benefits of the project and the government’s costs to procure it (α_0 minus any tax collected on profits):

$$\text{VFM} = b_H + t(\alpha_0^* - K + \delta e^*) - \alpha_0^* = b_H - K + \frac{\delta^2}{2}(1 - t^2)$$

If the project does change, the new contract price, α_1^* , will be given by, substituting (3) into (2):

$$\alpha_1^* = \frac{1 - \lambda}{1 - t}(b_H - b_L) + \frac{2\lambda - 1 - t}{1 - t}s + K - \frac{\delta^2}{2}(1 - t) \quad (4)$$

And the VFM after renegotiation will then be given by:

$$\text{VFM} = b_H + t(\alpha_1^* - K + \delta e^* - s) - \alpha_1^* - s = \lambda b_H + (1 - \lambda)b_L - K + \frac{\delta^2}{2}(1 - t^2) - 2\lambda s$$

Combining the VFM results with and without a change in project design, we see that the *ex ante* expected VFM is then:

$$\begin{aligned} E(\text{VFM}) &= \mu \left(\lambda b_H + (1 - \lambda) b_L - K + \frac{\delta^2}{2} (1 - t^2) - 2\lambda s \right) + (1 - \mu) \left(b_H - K + \frac{\delta^2}{2} (1 - t^2) \right) \\ &= (1 - \mu(1 - \lambda)) b_H + \mu(1 - \lambda) b_L - K + \frac{\delta^2}{2} (1 - t^2) - 2\mu\lambda s \end{aligned} \quad (5)$$

A set of intuitive results regarding the determinants of VFM of a project follow directly.

Result 1.²⁸ The value for money of a project will be greater: the lower is the cost of the project (K), the greater is the gross benefit of first- and second-best projects (b_H and b_L), the greater is the cost-reducing effect of effort (δ), the smaller is the probability (μ) the project design will need to change, the smaller is the switching cost (s), and the lower is the tax rate (t). Assuming that the net benefit of renegotiation ($b_H - b_L - 2s$) is always positive – that is, renegotiation is efficient and therefore always occurs when there are changes in demand – VFM is higher when the government is in a stronger bargaining position (i.e. when λ is greater).

Most of these results are easily understood and relate to the simple point that the greater the benefits and/or the lower are the costs of the project, the greater will be its VFM. The least obvious results here are those related to the tax rate, the bargaining weights and the likelihood of the need for renegotiation. By discouraging effort, higher tax rates actually hurt the achievement of VFM, even though the taxes received reduce government net expenditures. By limiting the ability of the private partner to drive up prices in renegotiation, a greater government bargaining weight increases expected VFM. Finally, the more likely it is that the project will need to be modified, the less attractive it becomes *ex ante* from a VFM perspective, even though the final gross benefits and costs of production will be in the same in any case.

Comparing PPPs and Traditional Public Procurement

We turn now to the key results of the paper: considering two special cases of the model above will allow us to compare the relative costs and efficiency of a PPP and more traditional methods of public procurement (PUB).

By taxing back all the profits (t approaching 1) and granting the government full power in renegotiation ($\lambda = 1$) we essentially transform the firm into an arm of the government and create the traditional public procurement scenario, PUB.²⁹ By contrast, when taxes are lower and bargaining weights more balanced, we have more private interests and control in the project.³⁰ To fix ideas here, we take the case in which $t = 0$ and $0 \leq \lambda \leq 1$ to represent the PPP model. Recall that for a given level of λ , lower taxes raise VFM, so indeed in a PPP scenario in which the government is ceding some control rights (i.e. $\lambda < 1$), the government will maximize VFM by setting the tax rate to zero.³¹

Setting $t = 0$ we can solve for the chosen level of effort, from (1), the original contract price, from (3), and the renegotiated price, from (4):